Inflation Persistence After an Energy-Price Shock: A Two-Sector New Keynesian Model with Production Networks and Real Wage Rigidity*

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Abstract

Abstract here.

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*COMMENTS. All remaining errors are my own.

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1 Introduction

The energy crisis in 2022 has led to a surge in natural gas price, which has in turn caused a significant increase in inflation in the UK. Whereas the natural gas price shock was transitory, the inflation response was lagged and persistent: the peak of the inflation response occurred a few months later than that of the natural gas price, and the inflation rate remained elevated for a long time after the natural gas price returned to normal. TODO: present observations!

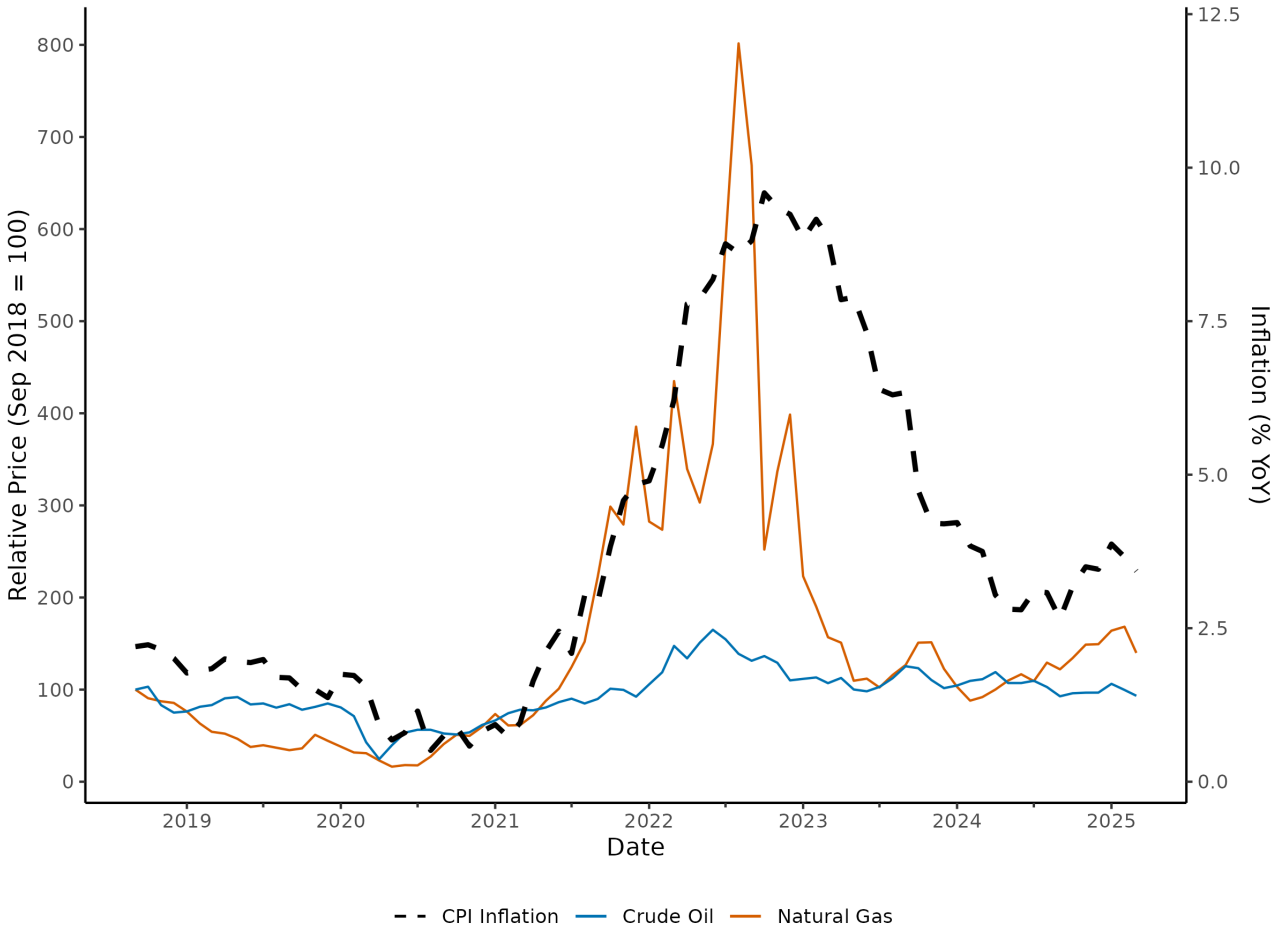

Figure 1: Energy Prices and UK CPI Inflation between September 2018 and March 2025

Notes: Solid lines show GBP-denominated natural gas (EU) and Brent crude oil prices indexed to September 2018 = 100 (left axis). Black dashed line shows headline CPI inflation, year-on-year (right axis). Sources: FRED (PNGASEUUSDM, DCOILBRENTEU, DEXUSUK, GBRCPALTT01CTGYM)

The standard New Keynesian model for a small open economy which imports energy as a factor of production at an exogenous price (hereafter "the baseline model"; see Appendix C for the log-linearised model equations) fails to replicate this hump-shaped and persistent response of inflation to the energy price shock.

I extend the baseline model in two following ways to replicate a hump-shaped and persistent inflation response to the energy price shock. First, I introduce production networks with an upstream industry that utilises imported energy and labour to produce intermediate goods and a downstream industry using the intermediate goods and labour to produce final

goods that are consumed by households. There is a single labour market for both industries, where workers move freely between the two industries. Consequently, the wage is identical across the two industries in the equilibrium. An important feature of this model's production network is that the downstream industry is myopic in its pricing decisions (see Minton and Wheaton (2023)). While non-myopic Calvo-pricing firms incorporate full information about the expected future marginal cost into their reset price calculations, myopic firms discount the future expected marginal cost and hence put more weight on the current marginal cost when setting their prices. The myopia of the downstream industry corresponds to the difficulty, particularly for downstream firms, to perfectly foresee the future path of energy price, compared to the upstream industry. TODO: more details here.

Second, I introduce real wage rigidity in the labour market, which represents the friction in the labour market such as fixed contracts, negotiation costs, and separation/hiring costs.

The rest of the paper is organised as follows. The following subsection explores the literature. In Section 2, I describe the baseline model and its extensions, as well as calibrate the model parameters for the simulation. In Section 3, I present the simulation results and evaluate them by conducting further sensitivity analysis. Finally, the paper concludes in Section 4.

Observation:

- Hump shaped response of inflation to oil price shock.
- Inflation rises after a lag, then falls with some persistence
- Production networks
  - Upstream industry: energy + labour -> intermediate goods
  - Downstream industry: intermediate goods + labour -> final goods
  - Myopic: can't perfectly foresee the future path of oil price, so they set prices (partly) based on current cost
- Real wage rigidity
  - representing the friction in the labour market, such as fixed contracts, negotiation costs, and separation/hiring costs
- Sensitivity analysis

1.1 Literature

Gagliardone and Gertler (2025) explores AAA.

Gomellini et al. (2025) explores BBB.

- Production networks
- Real wage rigidity
- Oil shock `https://www.nber.org/papers/w13368`

2 The Model

Plan: https://claude.ai/chat/72a44bc1-0451-4698-a8c1-b9c165442122

2.1 Economic Environment and Terms-of-Trade Shock

Consider a small open economy with two industrial sectors that imports a quantity Z_t of energy from the rest of the world. Assume the economy does not produce energy domestically, and its domestic demand for energy does not affect the global energy price. There is also a homogenous internationally tradable good that the economy can produce and export. Assume without loss of generality that the domestic economy can produce E_t units of this traded good using E_t units of domestic labour, and the market for the traded good is perfectly competitive. This implies that the domestic-currency price of the traded good is equal to the nominal wage W_t, the nominal marginal cost of production. Define the unit price of energy imports in domestic currency to be S_t, and denote the relative price of imports to exports (i.e., the terms of trade) as $s_t = S_t/W_t$. The small open economy takes this terms of trade s_t as given, and shocks to s_t represent energy price shocks in the analysis.

Assume that the terms of trade s_t follows an exogenous AR(1) process. Let $\hat{s}_t$ denote the log-deviation of s_t from its steady state value $\bar{s}$ which is normalised to one without loss of generality, so $\hat{s}_t$ follows

$$\hat{s}_t = \rho \hat{s}_{t-1} + u_t, \tag{1}$$

where u_t is a zero-mean i.i.d. shock to the terms of trade, and ρ measures the persistence of the exogenous shock.

All goods are perishable, and the model does not have financial assets to be traded internationally, so net exports must be zero in the balance-of-payments equilibrium. Mathematically, this means $W_t E_t = S_t Z_t$, or equivalently,

$$E_t = s_t Z_t. \tag{2}$$

2.2 Production Network

Apart from the production of tradable goods (which utilises E_t units of labour to produce the same quantity of the tradable goods), the economy entails a production network of domestic consumption goods, which consists of upstream and downstream industries. Both industries use domestic labour as an input, which does not require any industry-specific skills, and the workers are completely mobile between the two industries, so that the wage is identical across the two industries in the equilibrium.

In the upstream industry, there is a continuum of firms $j \in [0, 1]$ that uses imported energy and domestic labour to produce differentiated intermediate goods, which will be used as inputs by the downstream industry. The production of the upstream industry firm j

is described by the following Leontief production function:

$$Q_t(j) = \min\left\{\frac{Z_t(j)}{\alpha^U}, \frac{N_t^U(j)}{1-\alpha^U}\right\}, \tag{3}$$

where $Q_t(j)$ denotes the output of firm j in the upstream industry, $Z_t(j)$ and $N_t^U(j)$ denote the usage of energy and labour by firm j as inputs, and the parameter $\alpha^U \in (0,1)$ is the share of energy in the production function.[1] The production function is in Leontief form, which implies that there is no substitutability between energy and labour in producing the intermediate goods. The production cost is composed of the cost of labour and energy, whose nominal prices are W_t and S_t, respectively. As the upstream industry firm minimises its production cost, energy and labour are used in fixed proportions:

$$Z_t(j) = \alpha^U Q_t(j), \quad N_t^U(j) = (1-\alpha^U)Q_t(j). \tag{4}$$

The nominal marginal cost X_t^U of producing one unit of intermediate good is constant across firms in the upstream industry and is given by

$$X_t^U = \alpha^U S_t + (1-\alpha^U)W_t = (1-\alpha^U + \alpha^U s_t)W_t. \tag{5}$$

It follows that the real marginal cost $x_t^U = X_t^U/P_t^U$ is

$$x_t^U = \left(1-\alpha^U + \alpha^U s_t\right)\cdot\frac{W_t}{P_t^D}\cdot\frac{P_t^D}{P_t^U} = (1-\alpha^U + \alpha^U s_t)\cdot w_t \cdot \frac{P_t^D}{P_t^U}, \tag{6}$$

where $w_t = W_t/P_t^D$ is the real wage.[2] Since $\bar{s} = 1$ in steady state, the parameter α^U can be interpreted as the steady-state energy share of the upstream industry firm's production costs. The basket of differentiated intermediate goods is aggregated into a composite intermediate good using a CES aggregator:

$$Q_t = \left(\int_0^1 Q_t(j)^{\frac{\varepsilon-1}{\varepsilon}} dj\right)^{\frac{\varepsilon}{\varepsilon-1}} \tag{7}$$

which is supplied to the downstream industry to be used in the production of consumption goods. Here, ε is the elasticity of substitution within the upstream industry intermediate goods.[3]

In the downstream industry, there is a continuum of firms $i \in [0,1]$ that uses the composite intermediate goods produced by the upstream industry as well as domestic labour to

[1] I put a superscript U (D) to indicate that a variable is about the upstream (downstream) industry.

[2] Whereas measuring the real term of the upstream marginal cost by the price of intermediate goods P_t^U, I measure wage, which is constant across both industries, by the price of final consumption goods P_t^D.

[3] For simplicity, the model assumes the elasticity of substitution is the same for both upstream and downstream industries. I conduct sensitivity analysis on this assumption in Section 3.1 to show how results would change if it differs between the two industries.

produce differentiated final goods, which are consumed by households. The production of the downstream industry firm i is also in Leontief form:

$$Y_t(i) = \min\left\{\frac{M_t(i)}{\alpha^D}, \frac{N_t^D(i)}{1-\alpha^D}\right\}, \tag{8}$$

where $Y_t(i)$ denotes the output of firm i in the downstream industry, $M_t(i)$ and $N_t^D(i)$ denote the usage of intermediate goods composite and labour by firm i as inputs, and the parameter $\alpha^D \in (0,1)$ is the share of intermediate goods composite in the production function.[4] Similarly to the production of the upstream industry, the production cost minimisation implies that intermediate goods and labour are used in fixed proportions:

$$M_t(i) = \alpha^D Y_t(i), \quad N_t^D(i) = (1-\alpha^D)Y_t(i). \tag{9}$$

The nominal marginal cost X_t^D of producing one unit of the final consumption good is

$$X_t^D = \alpha^D P_t^U + (1-\alpha^D)W_t, \tag{10}$$

and the real marginal cost $x_t^D = X_t^D/P_t^D$ is

$$x_t^D = \left(1-\alpha^D + \alpha^D\frac{P_t^U}{W_t}\right)\frac{W_t}{P_t^D} = \left(1-\alpha^D + \alpha^D\frac{P_t^U}{W_t}\right)w_t. \tag{11}$$

Domestic households consume the CES aggregated basket of differentiated final goods:

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}}di\right)^{\frac{\varepsilon}{\varepsilon-1}}, \tag{12}$$

where $C_t(i) = Y_t(i)$ holds in the equilibrium for all i as the market clears. Solving the household's utility maximisation problem under the budget constraint yields the demand for each differentiated final good i: SHOULD I SOLVE EXPLICITLY?

$$Y_t(i) = \left(\frac{P_t^D(i)}{P_t^D}\right)^{-\varepsilon} Y_t, \tag{13}$$

where

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{\varepsilon-1}{\varepsilon}}di\right)^{\frac{\varepsilon}{\varepsilon-1}}, \quad P_t^D = \left(\int_0^1 P_t^D(i)^{1-\varepsilon}di\right)^{\frac{1}{1-\varepsilon}}. \tag{14}$$

Similarly, solutions to the downstream industry's cost minimisation problem yield the de-

[4] α^D can be interpreted as the (D,U)-element of the input-output matrix in a more generalised production network model.

mand for each intermediate good j by the downstream industry:

$$M_t(j) = \left(\frac{P_t^U(j)}{P_t^U}\right)^{-\varepsilon} M_t, \tag{15}$$

where $M_t = \int_0^1 M_t(i)di$ is the downstream industry's aggregate demand for the intermediate goods basket. In the equilibrium, the market for intermediate goods clears for each differentiated intermediate good j, so $M_t(j) = Q_t(j)$ for all j.

Since net exports are zero and there is no investment or government expenditure in the model, aggregate output of the downstream industry Y_t is equal to the real GDP, as well as aggregate consumption C_t in the equilibrium. Since consumers only consume final goods and do not directly consume intermediate goods or energy, CPI inflation is determined solely by the price level of final goods:[5]

$$\pi_t = \pi_t^D = \frac{P_t^D - P_{t-1}^D}{P_{t-1}^D}. \tag{16}$$

2.3 Market Clearing and Aggregation

Let Z_t denote the aggregate demand for energy by the downstream industry and N_t^D and N_t^U denote the aggregate demand for labour by the downstream and upstream industry, respectively:

$$Z_t = \int_0^1 Z_t(j)dj, \quad N_t^D = \int_0^1 N_t^D(i)di, \quad N_t^U = \int_0^1 N_t^U(j)dj. \tag{17}$$

Then, the total demand for labour in the economy L_t is the sum of the labour demand by the two industries as well as the competitive firms producing the tradable goods:

$$L_t = N_t^D + N_t^U + E_t. \tag{18}$$

By substituting the demand functions for intermediate (15) and final goods (13), one can obtain the aggregate demands for the intermediate goods M_t and labour N_t^D for the downstream industry and energy Z_t and labour N_t^U for the upstream industry: MAYBE MOVE

[5]For this reason, I hereafter drop the superscript D to emphasise the dynamics of the entire economy, while keep it for describing the downstream industry specifically, but they are the same.

THE INTERIM DERIVATIONS TO THE APPENDIX?

$$M_t = \int_0^1 M_t(i)di = \int_0^1 \alpha^D Y_t(i)di = \alpha^D \int_0^1 \left(\frac{P_t^D(i)}{P_t^D}\right)^{-\varepsilon} Y_t di \tag{19}$$

$$= \alpha^D \Delta_t^D Y_t \tag{20}$$

$$N_t^D = \int_0^1 N_t^D(i)di = \int_0^1 (1-\alpha^D)Y_t(i)di = (1-\alpha^D)\int_0^1 \left(\frac{P_t^D(i)}{P_t^D}\right)^{-\varepsilon} Y_t di \tag{21}$$

$$= (1-\alpha^D)\Delta_t^D Y_t \tag{22}$$

$$Z_t = \int_0^1 Z_t(j)dj = \int_0^1 \alpha^U Q_t(j)dj = \alpha^U \int_0^1 \left(\frac{P_t^U(j)}{P_t^U}\right)^{-\varepsilon} Q_t dj \tag{23}$$

$$= \alpha^U \Delta_t^U Q_t \tag{24}$$

$$N_t^U = \int_0^1 N_t^U(j)dj = \int_0^1 (1-\alpha^U)Q_t(j)dj = (1-\alpha^U)\int_0^1 \left(\frac{P_t^U(j)}{P_t^U}\right)^{-\varepsilon} Q_t dj \tag{25}$$

$$= (1-\alpha^U)\Delta_t^U Q_t, \tag{26}$$

where $\Delta_t^D = \int_0^1 \left(P_t^D(i)/P_t^D\right)^{-\varepsilon} di$ and $\Delta_t^U = \int_0^1 \left(P_t^U(j)/P_t^U\right)^{-\varepsilon} dj$ are the usual price dispersion terms for the downstream and upstream industry, respectively. Substituting the balance of payments condition (2) into the labour demand equation (18) gives $L_t = N_t^D + N_t^U + s_t Z_t$. Further substituting the aggregate demands derived above yields:

$$L_t = N_t^D + N_t^U + s_t Z_t \tag{27}$$

$$= (1-\alpha^D)\Delta_t^D Y_t + (1-\alpha^U)\Delta_t^U Q_t + s_t \alpha^U \Delta_t^U Q_t \tag{28}$$

$$= (1-\alpha^D)\Delta_t^D Y_t + (1-\alpha^U)\Delta_t^U Q_t + \alpha^U s_t \Delta_t^U Q_t \tag{29}$$

$$= (1-\alpha^D)\Delta_t^D Y_t + (1-\alpha^U+\alpha^U s_t)\Delta_t^U \alpha^D \Delta_t^D Y_t \tag{30}$$

$$= [1-\alpha^D + (1-\alpha^U+\alpha^U s_t)\alpha^D \Delta_t^U]\Delta_t^D Y_t, \tag{31}$$

which can be rearranged to express the aggregate output Y_t as a function of the total labour demand L_t and the terms of trade s_t:

$$Y_t = \frac{L_t}{[1-\alpha^D + (1-\alpha^U+\alpha^U s_t)\alpha^D \Delta_t^U]\Delta_t^D}. \tag{32}$$

This implies that the economy's effective labour productivity is

$$\frac{Y_t}{L_t} = \frac{1}{[1-\alpha^D + (1-\alpha^U+\alpha^U s_t)\alpha^D \Delta_t^U]\Delta_t^D}, \tag{33}$$

which declines as either or both of the industry-wise relative-price distortions Δ_t^D and Δ_t^U increase, and as the terms of trade s_t depreciates (i.e., imported energy price increases in domestic currency). Mathematically, the relative-price distortion in final goods Δ_t^D has a

larger negative effect on the economy's effective labour productivity than that in intermediate goods Δ_t^U, which is intuitive because the former directly affects the price of final goods, while the latter only indirectly affects it through the cost of intermediate goods. When the terms of trade s_t depreciates, this means that the relative price of energy imports increases compared to other traded goods, which will require more labour input to produce the tradable goods E_t to maintain the balance of payments equilibrium. Consequently, given the fixed amount of total labour in the economy L_t, less labour can be allocated to the production of intermediate and final consumption goods, which is equivalent to a decline in the economy's effective labour productivity. WHY NEED TO EXPLAIN THIS?

2.4 Households and Labour Supply

As per the standard New Keynesian framework, households with CRRA utility from consumption and disutility from labour supply maximise the expected discounted sum of utility subject to the budget constraint:

$$\max_{\{C_t, B_t, L_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-1/\sigma}}{1-1/\sigma} - \frac{L_t^{1+1/\psi}}{1+1/\psi} \right] \tag{34}$$

$$\text{s.t. } P_t^D C_t + B_t = (1+i_{t-1})B_{t-1} + W_t L_t + \Pi_t, \tag{35}$$

where β, σ, and ψ are the parameters of discount factor, elasticity of intertemporal substitution, and Frisch elasticity of labour supply, respectively, while B_t is the holding of nominal one-period bond, i_t is the nominal interest rate, and Π_t is the nominal profit from owning firms. FOCs with respect to C_t and L_t yield the Euler equation and the labour supply equation, respectively:

$$\frac{C_t^{-1/\sigma}}{P_t^D} = \beta(1+i_t)\mathbb{E}_t \left[\frac{C_{t+1}^{-1/\sigma}}{P_{t+1}^D} \right] \tag{36}$$

$$\frac{L_t^{1/\psi}}{C_t^{-1/\sigma}} = \frac{W_t}{P_t^D} = w_t. \tag{37}$$

ADD INTUITION FOR EULER EQN? The labour supply equation (37) implies that the real wage w_t equals the marginal rate of substitution between consumption and leisure when there is no friction in the labour market. The following subsection introduces real wage rigidity, however, which potentially generates a wedge between the real wage and the marginal rate of substitution.

2.5 Real Wage Rigidity

I introduce real wage rigidity in the labour market, which prevents the real wage from adjusting immediately to the labour market conditions. Mathematically, following Blanchard

and Galí (2007), I assume that the actual real wage w_t is a weighted geometric mean of the lagged real wage w_{t-1} and the flexible real wage w_t^f:

$$w_t = w_{t-1}^{\psi_w} \cdot (w_t^f)^{1-\psi_w}, \tag{38}$$

where $\psi_w \in [0, 1]$ is the parameter that governs the degree of real wage rigidity with $\psi_w = 0$ corresponding to the perfectly flexible labour market with no real wage rigidity.

The real wage rigidity reflects the friction in the labour market, such as fixed contracts, negotiation costs, and separation and hiring costs.[6]

2.6 Calvo and Myopic Price Setting

Both upstream and downstream firms adjust their individual nominal prices $P_t^U(j)$ and $P_t^D(i)$ according to the Calvo pricing model with probability θ_U and θ_D of no adjustment, respectively. The upstream industry firms determine their reset price P_t^{U*} by solving the standard expected discounted profit maximisation problem:

$$\max_{P_t^{U*}} \mathbb{E}_t \sum_{\ell=0}^{\infty} (\theta^U)^\ell q_{t+\ell|t} \left[\left(\frac{P_t^{U*}}{P_{t+\ell}^U} \right)^{1-\varepsilon} - x_{t+\ell}^U \left(\frac{P_t^{U*}}{P_{t+\ell}^U} \right)^{-\varepsilon} \right] Q_{t+\ell} \tag{39}$$

where $q_t = Q_t P_{t+1}^D / P_t^D$, $q_{t+\ell|t} \equiv q_t \cdots q_{t+\ell-1}$ is the real discounted factor.[7] Take the FOC with respect to P_t^{U*}:

$$\mathbb{E}_t \sum_{\ell=0}^{\infty} (\theta^U)^\ell q_{t+\ell|t} \left[(1-\varepsilon) \left(\frac{P_t^{U*}}{P_{t+\ell}^U} \right)^{-\varepsilon} + \varepsilon x_{t+\ell}^U \left(\frac{P_t^{U*}}{P_{t+\ell}^U} \right)^{-\varepsilon-1} \right] \frac{Q_{t+\ell}}{P_{t+\ell}^U} = 0, \tag{40}$$

which can be arranged as

$$P_t^{U*} = \frac{\varepsilon}{\varepsilon - 1} \cdot \frac{\mathbb{E}_t \sum_{\ell=0}^{\infty} (\theta^U)^\ell q_{t+\ell|t} x_{t+\ell}^U (P_{t+\ell}^U)^\varepsilon Q_{t+\ell}}{\mathbb{E}_t \sum_{\ell=0}^{\infty} (\theta^U)^\ell q_{t+\ell|t} (P_{t+\ell}^U)^{\varepsilon-1} Q_{t+\ell}}. \tag{41}$$

In words, the optimal reset price for the intermediate good is a markup over a weighted average of the expected future real marginal cost, where the weights are determined by the Calvo parameter θ^U, the real discount factor $q_{t+\ell|t}$, and the future output $Q_{t+\ell}$. ADD INTUITION? It follows that the upstream industry price level P_t^U is a weighted average of the lagged price level P_{t-1}^U and the reset price P_t^{U*}:

$$P_t^U = [\theta^U P_{t-1}^U + (1-\theta^U) P_t^{U*}]^{\frac{1}{1-\varepsilon}}. \tag{42}$$

[6] Though out of scope of the paper, microfoundation can be provided by, for example, introducing search and matching mechanism with Nash bargaining wage negotiation as discussed by Hall (2005).

[7] As a feature of Leontief production function, the marginal cost does not depend on the firm's own output, so the amount of produced intermediate goods $Q_{t+\ell}$ appears linearly in the objective function.

The downstream industry firms also adopt the same Calvo pricing model with a parameter θ^D for the probability of no price adjustment. Additionally, I allow deviations from rational expectations à la Gabaix (2020) and Minton and Wheaton (2023); the downstream firms are "myopic" in a sense that they discount the future expected marginal cost when setting their reset price, and hence put more weight on the current marginal cost than non-myopic firms do. To implement myopia, I simply replace the (rational) expectation operator $\mathbb{E}_t[\cdot]$ in the downstream firms' reset price setting problem with a *myopic expectation operator* $\tilde{\mathbb{E}}[\cdot]$ whose definition is given in the following subsection (since it is defined in terms of the log-deviation of variables from steady state):

$$\max_{P_t^{D*}} \tilde{\mathbb{E}}_t \sum_{\ell=0}^{\infty} (\theta^D)^\ell q_{t+\ell|t} \left[\left(\frac{P_t^{D*}}{P_{t+\ell}^D} \right)^{1-\varepsilon} - x_{t+\ell}^D \left(\frac{P_t^{D*}}{P_{t+\ell}^D} \right)^{-\varepsilon} \right] \frac{Y_{t+\ell}}{P_{t+\ell}^D}. \tag{43}$$

Except for the myopia, the FOC and the optimal reset price for the downstream industry are derived in the same way as the upstream industry.

2.7 Competitive Equilibrium

The model is closed by specifying a Taylor monetary policy rule, which I describe in the next subsection in the log-linearised form. Here, I formally define the competitive equilibrium of the model.

Definition (Competitive Equilibrium)**.** *A competitive equilibrium consists of prices* $\{P_t^D(i), P_t^U(j), W_t, i_t\}_{t=0}^{\infty}$ *and allocations* $\{C_t, Y_t(i), Q_t(j), L_t, N_t^D(i), N_t^U(j), Z_t, E_t, B_t\}_{t=0}^{\infty}$ *such that:*

1. *Households maximise utility with respect to consumption and bond holdings (i.e., the Euler equation (36) holds), while the real wage follows the partial adjustment process (38), therefore employment is determined by firms' labour demand at the prevailing real wage* w_t*.*

2. *The upstream industry firms set prices optimally under Calvo with parameter* θ^U*, whereas the downstream industry firms set prices optimally under Calvo with parameter* θ^D *and myopic expectations.*

3. *The labour market clears at the nominal wage* W_t *with aggregate labour supply* $L_t = N_t^D + N_t^U + E_t$*.*

4. *The intermediate goods market clears in aggregate* $Q_t = M_t$*, where* Q_t *is aggregate upstream output and* $M_t = \alpha^D \Delta_t^D Y_t$ *is aggregate demand for intermediate goods composite by the downstream industry.*

5. *The consumption goods market clears:* $Y_t = C_t$*.*

6. *The balance of payments equilibrium holds:* $E_t = s_t Z_t$*.*

7. *The bond market clears:* $B_t = 0$.

8. *The terms of trade* s_t *follows the exogenous AR(1) process (1).*

Model ignores changes in TFP either exogenous or endogenous by the energy price shock.

2.8 Log-linearisation

In this subsection, I log-linearise the model around the steady state. Detailed derivation is provided in Appendix A. 14 endogenous variables and 14 equations.

First, I define the myopic expectation operator $\tilde{\mathbb{E}}_t[\cdot]$ for the downstream firms's price setting problem.

Definition (Myopic Expectation). *Let* $\hat{v}_t$ *be the log-deviation of a variable* v_t *from its steady state value* $\bar{v}$*, i.e.,* $\hat{v}_t = \log(v_t/\bar{v})$*. The myopic expectation of* $\hat{v}_{t+k}$ *for* $k \geq 0$*, denoted by* $\tilde{\mathbb{E}}_t[\hat{v}_{t+k}]$*, is defined as*

$$\tilde{\mathbb{E}}_t[\hat{v}_{t+k}] = m_f^k \mathbb{E}_t[\hat{v}_{t+k}], \tag{44}$$

where $m_f \in [0, 1]$ *is the myopia parameter with* $m_f = 1$ *corresponding to the rational expectation case with no myopia, while* $m_f = 0$ *corresponding to the fully myopic case with no foresight at all.*

Second, the model requires a monetary policy rule to be closed. I use a standard log-linearised Taylor rule

$$\hat{i}_t = \phi \hat{i}_{t-1} + (1 - \phi)(\chi_\pi \hat{\pi}_t + \chi_y \hat{y}_t) \tag{45}$$

with parameters ϕ, χ_π, and χ_y.

Combined with the log-linearised Taylor rule and the exogenous AR(1) process for the terms of trade (1), the following equations complete the log-linearisation of the model:

$$\hat{y}_t = \hat{L}_t - \alpha^U \alpha^D \hat{s}_t \quad \text{(aggregate production function)} \tag{46}$$

$$\hat{x}_t^D = \Phi_{DU}(\hat{p}_t^U - \hat{p}_t^D) + (1 - \Phi_{DU})\hat{w}_t \quad \text{(downstream real marginal cost)} \tag{47}$$

$$\hat{X}_t^D = \hat{x}_t^D + \hat{p}_t^D \quad \text{(downstream nominal marginal cost)} \tag{48}$$

$$\hat{X}_t^U = \alpha^U \hat{s}_t + \hat{w}_t + \hat{p}_t^D \quad \text{(upstream nominal marginal cost)} \tag{49}$$

$$\hat{w}_t^f = \frac{1}{\sigma}\hat{y}_t + \frac{1}{\psi}\hat{L}_t \quad \text{(flexible real wage)} \tag{50}$$

$$\hat{w}_t = \psi_w \hat{w}_{t-1} + (1 - \psi_w)\hat{w}_t^f \quad \text{(real wage rigidity)} \tag{51}$$

$$\hat{\pi}_t = \hat{p}_t^D - \hat{p}_{t-1}^D \quad \text{(CPI inflation)} \tag{52}$$

$$\hat{p}_t^D = \theta^D \hat{p}_{t-1}^D + (1 - \theta^D)\hat{p}_t^{D*} \quad \text{(downstream Calvo price level)} \tag{53}$$

$$\hat{p}_t^{D*} = \beta m_f \theta^D \mathbb{E}_t[\hat{p}_{t+1}^{D*}] + (1 - \beta m_f \theta^D)\hat{X}_t^D \quad \text{(downstream reset price)} \tag{54}$$

$$\hat{p}_t^U = \theta^U \hat{p}_{t-1}^U + (1 - \theta^U)\hat{p}_t^{U*} \quad \text{(upstream Calvo price level)} \tag{55}$$

$$\hat{p}_t^{U*} = \beta \theta^U \mathbb{E}_t[\hat{p}_{t+1}^{U*}] + (1 - \beta\theta^U)\hat{X}_t^U \quad \text{(upstream reset price)} \tag{56}$$

$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \sigma(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]). \quad \text{(IS equation)} \tag{57}$$

Here, I derive the log-linearised equation for the downstream real marginal cost (47) to show the derivation of the internally calibrated parameter Φ_{DU}. Before log-linearisation, the downstream real marginal cost is given by

$$x_t^D = \left(1 - \alpha^D + \alpha^D \frac{P_t^U}{W_t}\right) w_t. \tag{11}$$

In the steady state, the nominal marginal cost of the upstream industry is $\bar{X}^U = \bar{W}$ from (5). Also, every firm in the upstream industry will set the optimal reset price, which is the steady state nominal marginal cost multiplied by the constant markup:

$$\bar{P}^U = \mu \bar{W} = \frac{\varepsilon}{\varepsilon - 1} \bar{W}, \tag{58}$$

where $\mu \equiv \varepsilon/(\varepsilon - 1)$ is the markup. It follows that the steady state real marginal cost of the downstream industry is

$$\bar{x}^D = (1 - \alpha^D + \alpha^D \mu)\bar{w}. \tag{59}$$

The log deviation of P_t^U/W_t is $\hat{P}_t^U - \hat{W}_t$, and since $w_t = W_t/P_t^D$, $\hat{W}_t = \hat{w}_t + \hat{p}_t^D$. Therefore,

$$\widehat{\left(\frac{P_t^U}{W_t}\right)} = \hat{p}_t^U - \hat{w}_t - \hat{p}_t^D. \tag{60}$$

Log-linearising x_t^D around the steady state gives

$$\begin{aligned}
\hat{x}_t^D &= \frac{\alpha^D \cdot \bar{P}^U/\bar{W}}{1 - \alpha^D + \alpha^D \cdot \bar{P}^U/\bar{W}} (\hat{p}_t^U - \hat{w}_t - \hat{p}_t^D) + \hat{w}_t && (61)\\
&= \frac{\alpha^D \mu}{1 - \alpha^D + \alpha^D \mu} (\hat{p}_t^U - \hat{w}_t - \hat{p}_t^D) + \hat{w}_t \qquad \text{since } \mu = \bar{P}^U/\bar{W} && (62)\\
&= \Phi_{DU}(\hat{p}_t^U - \hat{w}_t - \hat{p}_t^D) + \hat{w}_t, && (63)\\
&= \Phi_{DU}(\hat{p}_t^U - \hat{p}_t^D) + (1 - \Phi_{DU})\hat{w}_t, && (64)
\end{aligned}$$

where

$$\Phi_{DU} = \frac{\alpha^D \mu}{1 - \alpha^D + \alpha^D \mu}. \tag{65}$$

2.9 Calibration

I calibrate the model based on the UK economy around the 2022 Energy Crisis. The model period is monthly. Summary of the parameter values is presented in Table 1. Here, I describe the sources and rationales of the calibration choices for each parameter. I provide sensitivity analysis in Section 3.1 to show how the results change if different parameter values are chosen.

First, the three parameters in the households' utility function follow the standard in the

New Keynesian Literature: the discount factor β is set to 0.9984, implied by the annual real interest rate of 2%; elasticity of intertemporal substitution σ is set to 0.5; and the Frisch elasticity of labour supply ψ is set to 3.

In the production network, I set the elasticity of substitution ε to 11, which implies the standard New Keynesian steady state markup of $\mu = 1.1$. Meanwhile, I set the energy cost share in the upstream industry α^U to 0.375. The upstream industry in the model represents energy-intensive sectors that utilise imported energy as input to produce intermediate goods. Using the annual Supply and Use Tables from the ONS, I calculated the output-weighted average energy cost share across the eight most energy-intensive and intermediate-goods-producing sectors annually from 2017 to 2022.[8] I adopt the average of the annual figures for 2017-2019 as the steady state value for $\alpha^U = 0.375$. α^D indicates the intermediate input share in the downstream industry, which is set to be 0.4. This implies that the overall energy cost share in the final consumption goods is $\alpha^U \alpha^D = 0.15$. Φ_{DU} is derived as

$$\Phi_{DU} = \frac{\alpha^D \mu}{1 - \alpha^D + \alpha^D \mu} = 0.4231, \tag{66}$$

which is the cost-based share of the intermediate goods in the downstream industry's real marginal cost. The difference to α^D is that the upstream industry's markup on the intermediate goods is taken into account in Φ_{DU}.

Calvo pricing parameters are set to be $\theta^D = 11/12$ and $\theta^U = 3/4$, which imply the average price durations of 12 months and 4 months for the downstream and upstream industry, respectively. A strand of literature has estimated the price durations using micro-level price data and tested the validity of the Calvo pricing model. Using the US data, Nakamura and Steinsson (2008) find that the median duration of consumer price changes is 8-11 months excluding sales, while that of producer price changes for intermediate goods is 7.0 months, implying that prices for intermediate goods are more flexible than those for final consumption goods. ADD MORE? UK data: Bunn and Ellis (2012a), Bunn and Ellis (2012b), Zhou and Dixon (2019) The myopia parameter for the downstream industry $m_f = 0.5$ implies that the downstream firms behaviourally discount the future expected marginal cost by 50% when setting their reset price. Minton and Wheaton (2023) estimate the myopia parameter for firms vis-à-vis shocks in oil prices and non-oil commodity prices. They conclude that downstream firms are almost rational ($m_f \approx 1$) with respect to oil price shocks, while almost completely myopic ($m_f \approx 0$) in case of price changes in non-oil commodities, potentially due to the difficulty in formulating forward-looking expectations on far end of supply chains. In my model, upstream firms produce a variety of intermediate goods using imported energy, of which downstream firms may or may not be able to easily forecast future price changes. Therefore,

[8] Specifically, the eight sectors are Electric Power Generation (`CPA_D351`), Gas Manufacture & Distribution (`CPA_D352_3`), Industrial Gases, Inorganics, Fertilisers (`CPA_C20A`), Petrochemicals (`CPA_C20B`), Paper and Paper Products (`CPA_C17`), Glass, Ceramics, Stone (`CPA_C23OTHER`), Basic Iron and Steel (`CPA_C241_3`), and Other Basic Metals & Casting (`CPA_C244_5`). Detailed calculations are available in Appendix B. Limitation: those industries also produce final goods.

for the main result, I adopt the midpoint of the possible value range $m_f = 0.5$ and provide sensitivity analysis with other values, $m_f \in \{0, 0.25, 1\}$.

Regarding the real wage rigidity parameter ψ_w, I adopt the monthly equivalent value $0.9^{1/3} \approx 0.965$ from the baseline calibration value (0.9 in quarterly frequency) in Blanchard and Galí (2007).

The monetary policy Taylor rule equation has three parameters, for which I set as $\phi = 0.9, \chi_\pi = 3$, and $\chi_y = 0.5$ following the literature standards with Taylor principle being satisfied ($\chi_\pi > 1$). ADD MORE?

Finally, ρ indicates the degree of persistence of the terms of trade $\hat{s}_t$, which follows an AR(1) process. I set this to $\rho = 0.8$, which implies that the size of an exogenous shock to the terms of trade decays half within 3.1 months.

Table 1: Model Parameter Values

Parameter	Symbol	Value
Households		
Discount factor	β	0.9984
Elasticity of intertemporal substitution	σ	0.5
Frisch elasticity of labour supply	ψ	3
Production and cost structure		
Energy cost share (upstream)	α^U	0.375
Intermediate input share (downstream)	α^D	0.4
Elasticity of substitution	ε	11
Downstream upstream-cost share	Φ_{DU}	0.4231
Price setting		
Downstream Calvo probability	θ^D	11/12
Upstream Calvo probability	θ^U	3/4
Myopia parameter (downstream)	m_f	0.5
Labour market		
Real wage rigidity	ψ_w	$0.9^{1/3} \approx 0.965$
Monetary policy		
Interest-rate smoothing	ϕ	0.9
Inflation response	χ_π	3
Output response	χ_y	0.5
Shock process		
Terms of trade persistence	ρ	0.8

https://www.ons.gov.uk/economy/nationalaccounts/supplyandusetables/datasets/ukinputoutputanalyticaltablesdetailed UK input-output tables, potentially useful to calibrate α^U and α^D.

3 Results and Discussion

Figures 2 and 3 display the model simulation results.

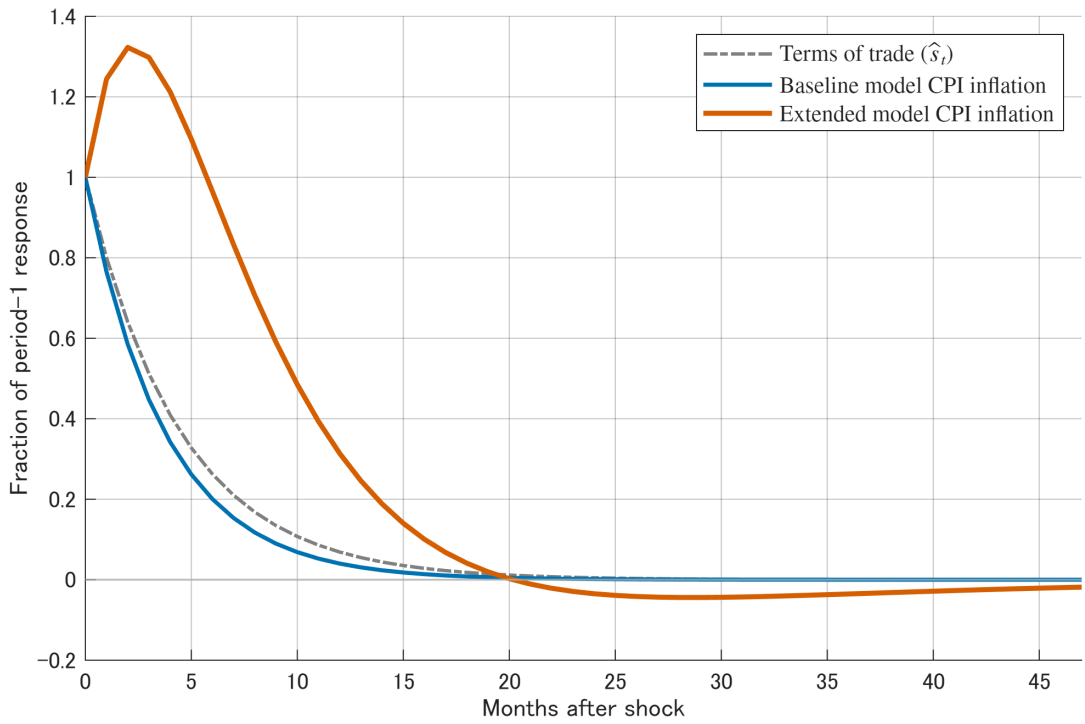

Figure 2: Inflation Persistence vs Energy Price Shock Decay

Notes:

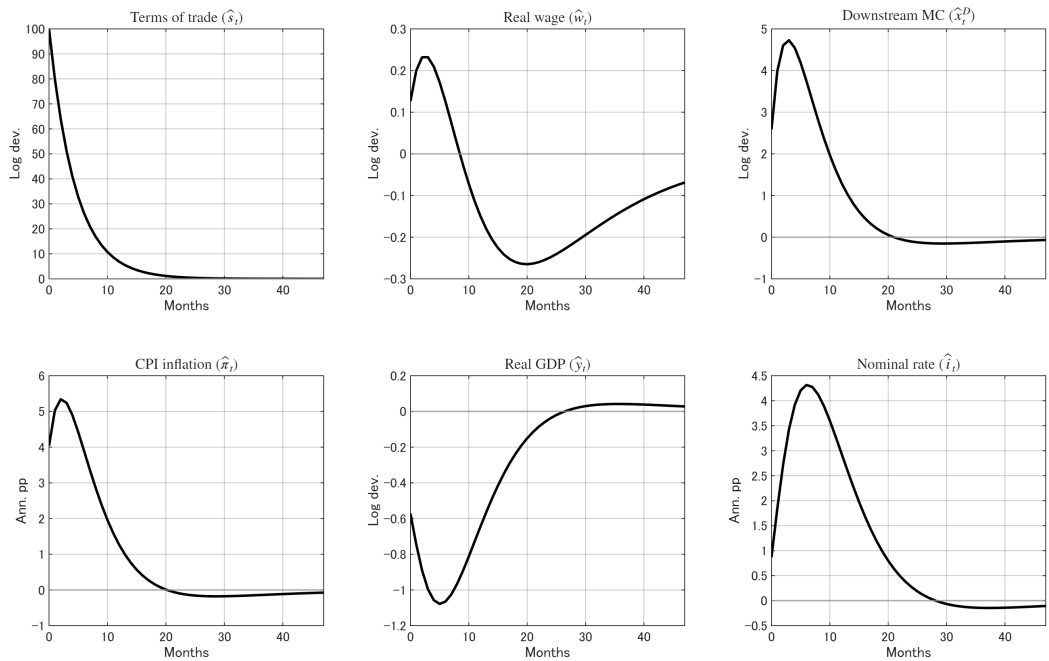

Figure 3: Impulse Responses of the Model to an Exogenous Terms of Trade Shock

Notes:

3.1 Sensitivity Analysis

Sensitivity analysis:

- Varying the degree of real wage rigidity $\psi_w \in \{0.705\}$
- upstream Calvo pricing parameter $\theta_U \in \{0.5\} + \{0, 0.3, 0.7\}$ (Nakamura and Steinsson, 2008)
- Downstream myopica parameter $m_f \in \{0, 0.5, 1\}$
- CES rather than Leontief production function?
- Values of α's

Policy implications?

4 Conclusion

Model limitations:

- Network structure of production and input-output linkages

Although the model assumes that the upstream industry firms imports energy by themselves, in reality, some firms in such industries may purchase energy from domestic energy retailers. However, in terms of price setting, it is still relevant since in data literature finds price of crude materials is almost completely flexible.

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A Appendix: Derivation of Log-linearised Equations

B Appendix: Calculations

Table? and graph for estimating α^U

C Appendix: The Baseline Model

This appendix section presents the baseline small open economy New Keynesian model, on which the paper's main model is extended. The baseline model is characterised by the following ten log-linearised equations:

$$\hat{y}_t = \hat{L}_t - \alpha \hat{s}_t \quad \text{(aggregate production function)} \tag{A.1}$$

$$\hat{x}_t = \hat{w}_t + \alpha \hat{s}_t \quad \text{(real marginal cost)} \tag{A.2}$$

$$\hat{X}_t = \hat{x}_t + \hat{P}_t \quad \text{(nominal marginal cost)} \tag{A.3}$$

$$\hat{w}_t = \frac{1}{\sigma}\hat{y}_t + \frac{1}{\psi}\hat{L}_t \quad \text{(labour supply)} \tag{A.4}$$

$$\hat{\pi}_t = \hat{P}_t - \hat{P}_{t-1} \quad \text{(inflation)} \tag{A.5}$$

$$\hat{P}_t = \theta \hat{P}_{t-1} + (1-\theta)\hat{P}^*_t \quad \text{(Calvo price level)} \tag{A.6}$$

$$\hat{P}^*_t = \beta\theta \mathbb{E}_t[\hat{P}^*_{t+1}] + (1-\beta\theta)\hat{X}_t \quad \text{(reset price)} \tag{A.7}$$

$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \sigma\big(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]\big) \quad \text{(IS equation)} \tag{A.8}$$

$$\hat{i}_t = \phi \hat{i}_{t-1} + (1-\phi)\big(\chi_\pi \hat{\pi}_t + \chi_y \hat{y}_t\big) \quad \text{(Taylor rule)} \tag{A.9}$$

$$\hat{s}_t = \rho \hat{s}_{t-1} + u_t \quad \text{(terms-of-trade shock)} \tag{A.10}$$

For calibration, the adopted parameter values are summarised in Table A.1.

Just list the log linearised equations here?

Table A.1: Baseline Model Parameter Values

Parameter	Symbol	Value
Households		
Discount factor	β	0.9984
Elasticity of intertemporal substitution	σ	0.5
Frisch elasticity of labour supply	ψ	3
Production and cost structure		
Energy cost share	α	0.15
Price setting		
Calvo probability	θ	11/12
Monetary policy		
Interest-rate smoothing	ϕ	0.9
Inflation response	χ_π	3
Output response	χ_y	0.5
Shock process		
Terms of trade persistence	ρ	0.8